Energy Efficiency Optimization using IoT-Driven Power Consumption Analytics

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Introduction

- Modern households and industries face rising electricity usage due to inefficient appliance behavior, poor demand prediction, and lack of real-time monitoring.
- This project aims to address these issues by combining IoT sensors with Machine Learning models to monitor, analyze, and forecast energy consumption.
- The system helps users optimize their energy usage by providing future load predictions, detecting inefficient patterns, and offering smart recommendations.

Literature Review

- Existing works show the potential of time-series forecasting for predicting electricity demand.
- Research shows LSTM outperforms traditional models in capturing non-linear and temporal dependencies in energy datasets.
- IoT-based energy monitoring using sensors (PZEM-004T, CT sensors) provides real-time power and voltage data for smart systems.

Research Gaps

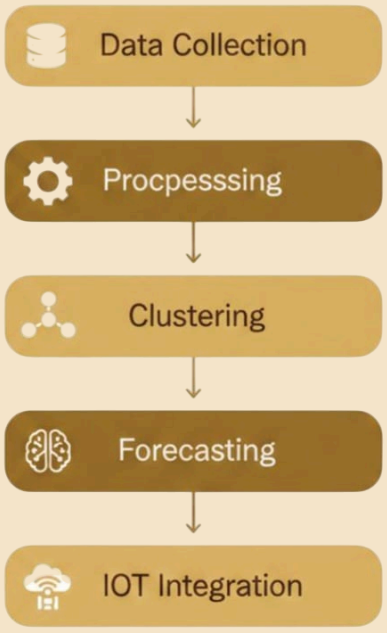
- Traditional forecasting methods fail to model complex energy consumption patterns.
- Lack of integrated systems combining real-time IoT data with ML forecasting.

Project Objectives

- Phase 1 – Machine Learning Model Development
  - To analyze historical energy consumption data and build ML models (Clustering + LSTM) for identifying inefficient usage patterns and forecasting future power demand.
- Phase 2 – IoT Integration & Real-Time Deployment
  - To integrate IoT energy sensors with the ML models to enable real-time monitoring, prediction, and smart recommendations for optimizing energy usage in households/industries.

Proposed Methodology

- Data Collection:Use historical dataset; later acquire real-time readings via IoT sensors.
- Preprocessing:Clean, resample, and engineer features (Power, kWh, PF, time patterns).
- Clustering:Apply K-Means to identify energy usage behaviour patterns.
- Forecasting:Train an LSTM model using past 24-hour sequences to predict future demand.
- IoT Integration:Stream live sensor data to ML model for real-time prediction and recommendations.



Identification of Tools/ Technology / Algorithms

- Tools & Technologies :
  - Python, Pandas, NumPy
  - Scikit-Learn, TensorFlow/Keras
  - ESP32, PZEM-004T, MQTT (for IoT integration)
- Algorithms :
  - K-Means Clustering – detect usage behavior patterns
  - LSTM (Long Short-Term Memory) – power demand forecasting

References

- UCI ML Repository – Individual Household Electric Power Consumption Dataset.
- IIETA's Paper on Time-Series Forecasting Models for Smart Meters Data: An Empirical Comparison and Analysis
- PZEM-004T & SCT-013 Sensor Datasheets.